

Social Media and Consumer Behavior Analysis

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The Design Choice

For this project, our team chose to implement a range of visualization approaches to effectively communicate insights about consumer behavior, especially in the context of social media's impact on purchasing decisions. We focused on a few key design principles to guide our decisions: clarity, interactivity, and a focus on actionable insights for marketing strategies.

1. Static Bar Chart

Description:

This chart displays the number of purchases (on the y-axis) for each product category (on the x-axis). The categories are arranged in a way that clearly shows the frequency of purchases across different product categories.

Rationale:

A static bar chart was chosen for its simplicity and effectiveness in conveying a clear comparison between product categories. This visualization helps users quickly identify which categories are the most popular in terms of social media-driven purchases.

Why This Design?

The bar chart allows for easy comparison of purchase frequencies, making it ideal for understanding trends and the dominance of specific categories like Electronics, Beauty, and Fashion across platforms. It provides a clear and simple way to identify top-performing categories and their relevance in the market.

2. Interactive Box Plot

Description:

The interactive box plot shows the amount spent (in USD) on the y-axis and product category on the x-axis. Users can select different social media platforms (Instagram, Twitter, etc.), different product categories (Fashion, Electronics, etc.), and customize the spending range (USD).

Rationale:

The box plot was chosen because it offers a concise representation of spending distribution, including the median, interquartile range, and potential outliers. The interactivity allows users to filter by platform and product category, providing a dynamic view of how spending varies across these factors.

Why This Design?

This visualization helps uncover the spread and central tendency of spending within each category, as well as how the spending behavior differs by social media platform. The interactive features allow users to drill deeper into the data and understand trends within specific contexts, such as platform-driven spending patterns.

3. Interactive Scatter Plot

Description:

The interactive scatter plot shows the amount spent (in USD) on the y-axis and product category on the x-axis. Users can filter by different social media platforms (Instagram, Twitter, etc.), product categories (Fashion, Electronics, etc.), and customize the spending range (USD).

Rationale:

We chose a scatter plot for this visualization because it is ideal for revealing correlations between variables. By adding interactivity, users can explore how spending varies across different social media platforms and product categories, which helps identify trends in consumer behavior.

Why This Design?

The scatter plot offers a detailed view of individual spending instances, which is especially useful for understanding outliers and the distribution of spending within each category. The ability to filter by platform and product category further enriches the user experience, making this a powerful tool for in-depth analysis.

4. Interactive Heatmap

Description:

The heatmap visualizes the relationship between income (USD annually) on the x-axis and amount spent (USD) on the y-axis. Users can interact with the plot by adjusting the city they would like to observe, such as New York, Los Angeles, etc.

Rationale:

A heatmap was selected because it can effectively show the intensity of spending across different income levels and cities. This visualization provides a clear view of how income and location influence spending behavior.

Why This Design?

The heatmap's color gradients make it easy to spot trends, such as areas where higher income individuals tend to spend more on social media. The city filter allows users to observe regional variations in spending, providing valuable insights for location-based marketing strategies.

5. Interactive Scatter Plot with Influence Level

Description:

This scatter plot shows the amount spent (in USD) on the y-axis and influence level on the x-axis. It includes interactive sliders to view specific influence levels, filter by income, and adjust the size of the points for readability.

Rationale:

A scatter plot was chosen because it is effective in showing the relationship between spending and influence level. The interactivity enhances user engagement, allowing for dynamic exploration of how different influence levels correlate with spending behavior.

Why This Design?

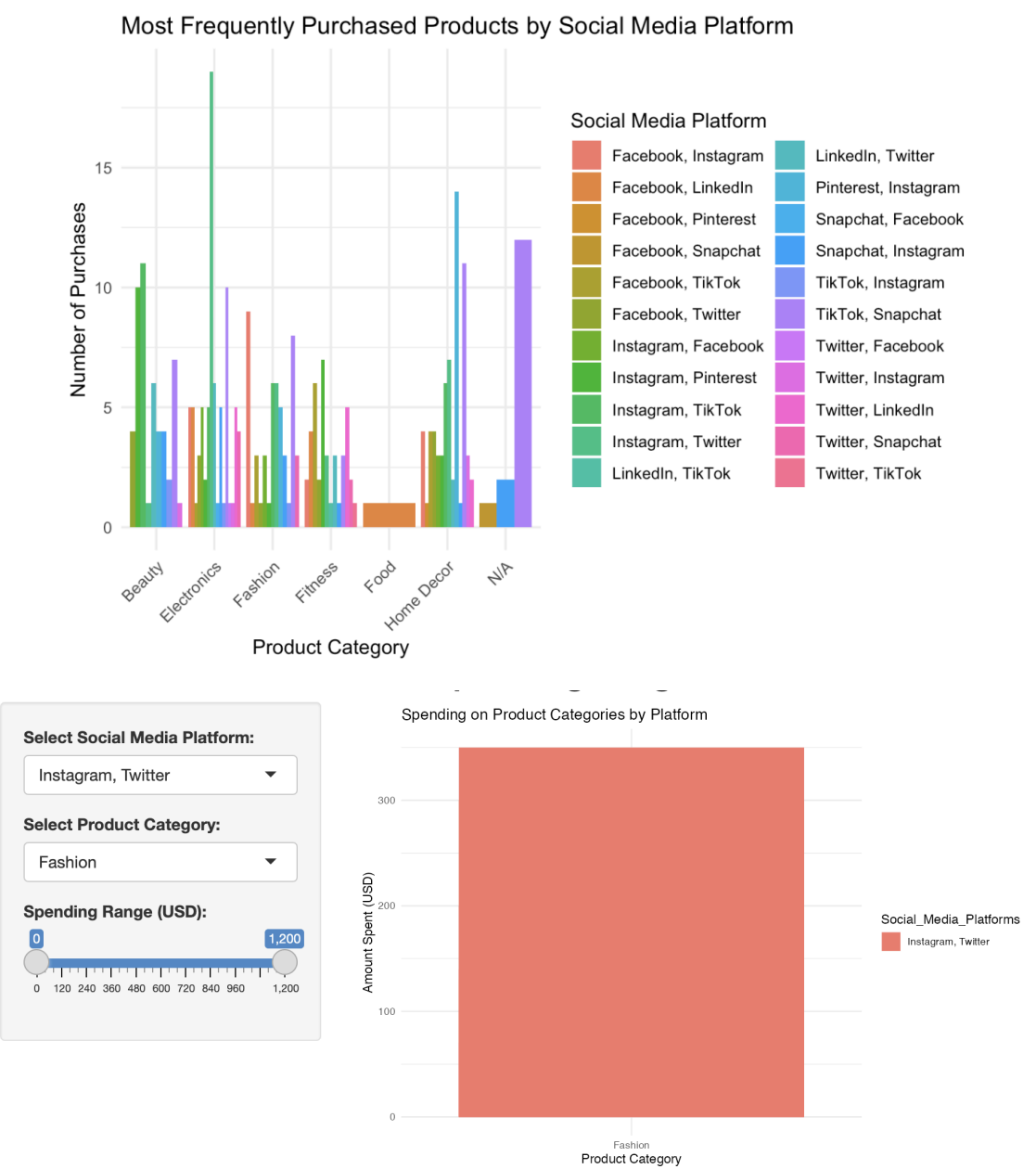
This visualization helps marketers understand how social media influence relates to spending. The sliders for income and influence level allow users to filter and zoom in on specific segments of the data, making it easier to identify high-value consumers or outliers. Adjusting the size of the points further improves readability, ensuring that key insights are easy to extract.

Conclusion:

Each of these visualization approaches was chosen to provide users with a clear, interactive, and actionable understanding of consumer behavior in relation to social media platforms. By incorporating interactive elements, we allow users to explore the

data from multiple angles, gaining deeper insights into trends across product categories, income levels, influence levels, and regional differences. These visualizations help marketers and business analysts make informed decisions based on consumer preferences, enabling them to craft targeted strategies that resonate with their audience.

Plot Analysis and Discussion



Based on the bar chart, Electronics emerges as the most frequently purchased product category, with the highest number of purchases across multiple social media platforms. This suggests that consumers are highly engaged with tech-related advertisements and influencer recommendations, particularly on platforms like Instagram, TikTok, and Facebook, which dominate in driving purchasing decisions. The strong presence of these platforms in Electronics purchases aligns with the trend of tech reviews and promotional content being widely shared on social media.

Beyond Electronics, Beauty and Fashion also see significant engagement, particularly on Instagram, TikTok, and Pinterest. The highly visual nature of these platforms makes them effective for showcasing products through influencers, brand promotions, and user-generated content. In contrast, Home Decor purchases are notably driven by Pinterest and Facebook, where lifestyle-focused content encourages consumers to explore new products. Fitness-related purchases show moderate engagement, likely influenced by the increasing popularity of health and wellness content on social media.

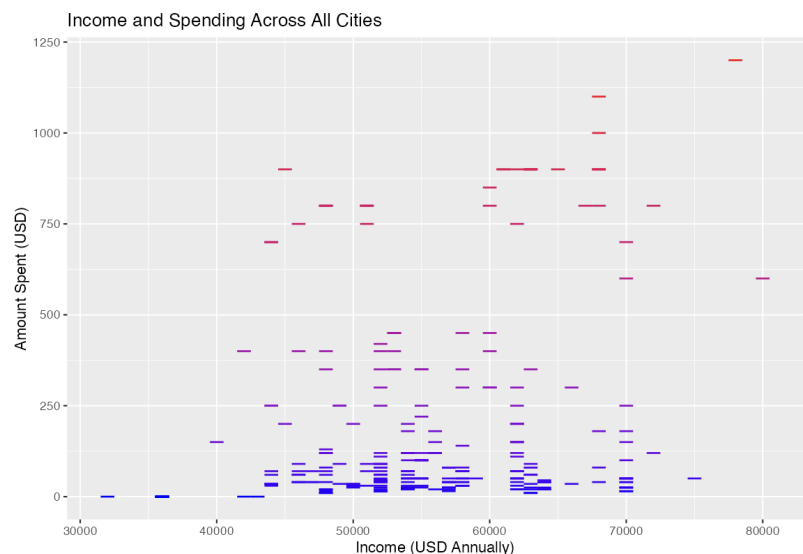
Income and Location Analysis

Select City:

All

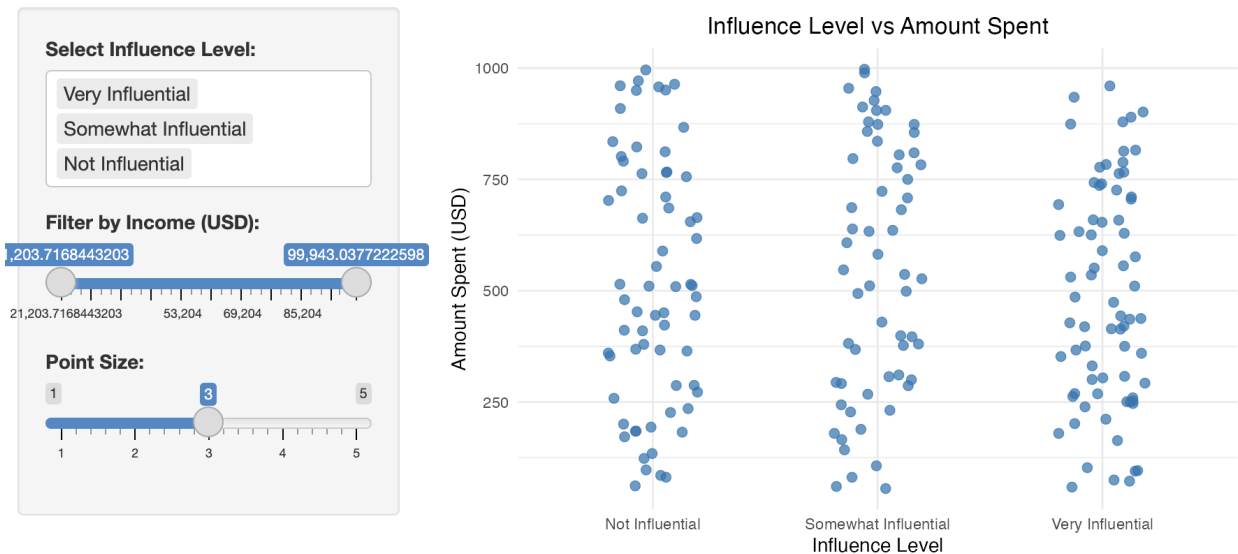
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Click to see details



This visualization shows the correlation between income and amount spent while on social media and is filterable by location of user. The most interesting finding of this graph is how variable the correlation is between locations, with some locations showing the highest purchases being from lower income and others showing the inverse or a lack of correlation. However, when looking at the full graph there is a definite noticeable trend. This leads to an interesting question of whether or not the culture of a city has more to do with social media spending than personal finances.

Consumer Spending Analysis



1. Spending Distribution Across Influence Levels

Not Influential:

Consumers in this category show a wide range of spending patterns. Some spend relatively small amounts, while others make high-value purchases. However, spending appears to cluster around the mid-range (\$200–\$600).

Somewhat Influential:

This group shows a similar spread but seems to have fewer consumers at the highest spending levels compared to the "Very Influential" group.

Very Influential:

This group has a notable concentration of consumers spending close to the maximum observed values (near \$1000). They appear to show higher willingness to make substantial purchases.

2. Spending Trends

While all groups exhibit a broad range of spending, the "Very Influential" group seems to have a higher density of consumers spending in the upper range. This suggests a possible trend where individuals with more social media influence are more willing to spend significantly on products.

3. Potential Outliers

Each influence category has a few individuals spending at the upper limit. These outliers may represent particularly engaged consumers or those with high disposable income.

4. Insights for Marketing Strategies

Targeting Strategy: Since "Very Influential" consumers tend to spend more, marketing campaigns that leverage social media influencers could be effective in driving higher-value sales.

Product Positioning: Since consumers across all influence levels show a broad spending range, brands may benefit from offering both premium and budget-friendly options to cater to diverse spending behaviors.

Critical Evaluation

Our Social Media Platform Engagement Heat Map revealed a fundamental issue that needs addressing. Particularly, our attempt to calculate means for categorical variables, which doesn't work statistically. This mistake is why our heatmap wasn't included in the final submission.

While our other five visualizations (product category bar chart, spending box plot, product category scatter plot, income-spending heatmap, and influence level scatter plot) work well together, this heatmap could still add value if fixed properly. The current version also has issues with its narrow color range and unclear labeling of what "Avg. Engagement" actually means. To fix this, we should either create a frequency-based heatmap showing counts instead of means, or switch to a bar chart that better handles categorical data. With these changes, we could add a useful sixth visualization to strengthen our overall analysis of social media's impact on consumer behavior.

Shiny Application



[Dataset](#)



[Shiny App](#)